

Daily Double optimal bet amounts in Jeopardy!

Abstract

Jeopardy! is a popular American trivia game show in which contestants are presented with clues in the form of answers and must provide the corresponding question. In this article, I discuss the importance of a strategic decision, the amount to wager on Daily Double, in order to maximize the chances of winning. I present a simulation model that can be used to determine the optimal wager for the final Daily Double based on the current board state and player scores. The results of the model show that players who are willing to take a calculated risk by betting more on Daily Doubles can increase their chances of winning by up to 19% over those that bet the average amount.

Intro

Jeopardy! is a trivia contest that pits three players against each other with a reversed "Answer" and "Question" format. It is estimated that of the 200,000 yearly applicants, only 400 contestants appear in a season. If you are one of the lucky few, then maximizing your chances of winning is paramount! I found myself in that position in 2020 - by wagering more aggressively than the "standard" Daily Double wager, I was able to increase my odds of winning.

The most famous literature around Jeopardy! is from IBM's Watson group [1]. Much of the novelty and discussion was around Watson's "understanding" of clues and how it parsed the answers in real time - matching or exceeding counterparts in gameplay thanks to advances in Natural Language Processing. Additional research covers the strategic decisions required to play the game [2]. They cover the four major strategic areas involved in a game of Jeopardy!: Daily Double wager amount, final jeopardy wager, clue selection, buzz in decision. I will be focusing on Daily Double wager amount as I believe most humans are too conservative in their wagers. Additionally, betting big on Daily Doubles was shown to be an effective strategy during James Holzhauer's run, often putting the game out of reach before Final Jeopardy. [3]

There are two leads a player can have going into Final Jeopardy: a standard lead ($P1_score > \max(P2_score, P3_score)$) and lock lead ($P1_score > 2 * \max(P2_score, P3_score)$) in which the player can't be caught if they bet \$0 on Final Jeopardy. Due to the uncertainty and typical higher difficulty of the Final Jeopardy question, it may be advantageous to play for a lock game in Double Jeopardy.

Implementation:

To initialize a game requires an array of player scores as well as an array containing the point values of all clues left on the board. The Daily Double wager amount is the variable we control. By running Monte Carlo simulations with different wager amounts, we can find which bets offer players the highest chance to win. For this simulation, I used my game from 2020-11-09 [4] in which I found the last Daily Double and wagered \$5000. The scores and the board state were as follows:

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scores = [19400, 6400, 17200]
board = [2000, 800, 1200, 1200, 400, 1600, 2000, 800, 800, 800, 400, 400, 400]
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The game proceeds by generating a Bernoulli variable with probability p , given by the players estimated confidence in their ability to answer the Daily Double correctly. If the value is $< p$, then the wager is “correct” and added to the players score. If the value is $\geq p$, the wager is “incorrect” and deducted from the players score. From here, the game continues by selecting additional clues (in the order given in the array) and adding their value to each player with equal probability ($\frac{1}{3}$). This assumes that all players are equally likely to buzz in and answer correctly. The game continues until all clues are selected. At this point, the results of the game are added to a series of results and the next trial is started.

We conduct trials by varying the wager (above) as well as the confidence in the answer. The confidence, while hard to estimate, is selected from a set of 5 values from 35% to 75%. This range includes the average % correct on Daily Doubles, with higher chances representing additional skill in the category or confidence from answering other questions correctly. From my own personal trivia statistics, I had a 55% chance of answering a History question correctly. My prior data analysis before going on the show suggested I had $< 50\%$ chance to answer correctly as it was a \$1600 value clue. The season data suggests a 61% chance of getting a \$1600 row Daily Double correct [5]. It is worth noting that this number is skewed high by returning champions often having control of the board and hunting for Daily Doubles.

Overall, there were 220 (5 confidence levels * 44 bet amounts) different samples to test. Over 10 million simulation runs resulted in ~45,000 runs for each sample. Through the central limit theorem, this allows us to converge on a result that is approximately normal.

$$\bar{x} = \frac{Normal(n\mu, \sqrt{n}\sigma)}{n} = Normal(\mu, \frac{\sigma}{\sqrt{n}})$$

Results:

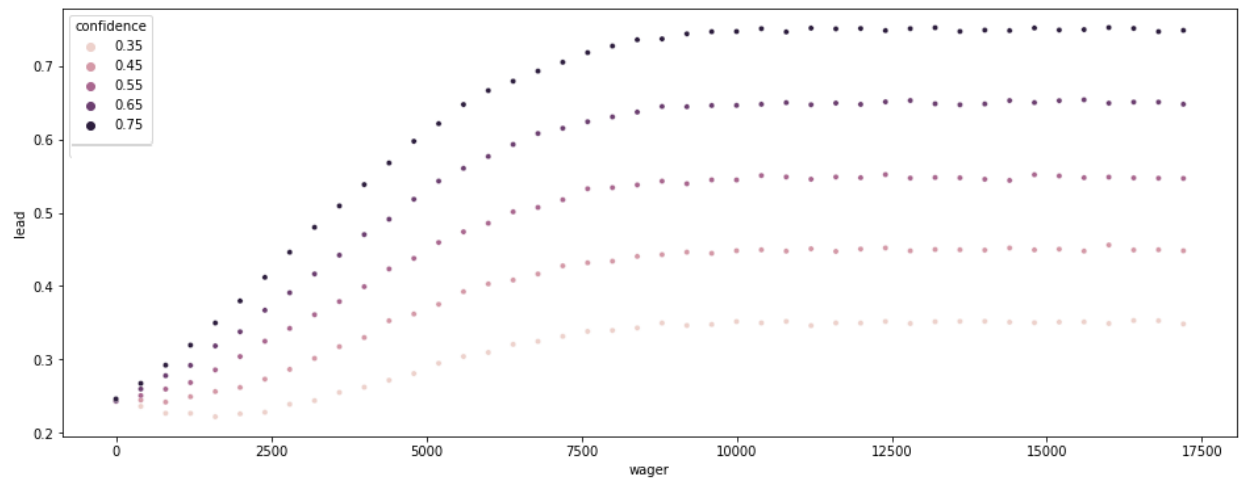


Fig 1. Wager vs. % of simulations where Daily Double player leads at end of round

From the results, I should have bet more! My win equity was 45% for confidence of 55% and a \$5000 wager. Interestingly, given this scenario, there was no confidence threshold between 35% and 75% right where I should not have bet a greater amount. Ultimately, my win equity would have increased by 10% had I bet correctly. However, the average Daily Double wager was \$3200 for season 37 [5]. The win equity for a confidence of 55% and wager of \$3200 is 36%. Compared to that, my wager improved the “standard” bet by 9%. Additionally, this is a simplistic view of what happens and it is not possible to figure out the exact probabilities of how often a game would reach this exact state due to uncertainty in player’s ability to answer questions.

Confidence	Wager	Win %
55%	3200	36%
55%	5000	45%
55%	12400	55%

Figure 2. Selected Win % given confidence level and wager amount

Further work:

There are numerous ways to add greater depth to the model - I believe simulating the game from the start and including clue selection as a decision within the game would help generalize play and wagers for the entire game. Involved in this is understanding the tradeoffs between trying to maximize score at the end of the Double Jeopardy round compared to winning Final Jeopardy, understanding under what parameters it better to go for lock game vs losing it all (for purely rational players as losing with \$0 might sting the ego!), and accounting for category knowledge when determining wager amounts. Much of this is covered in the follow up IBM

paper, but even they were limited in the number of simulations they could run. Backtesting on games is another interesting method to help understand how much skill or luck contributed to one's win. Given an existing game, replaying it with Daily Doubles placed in different areas would help players understand how well they played a game and how frequently they were aided by luck.

Works Cited

- [1] D. A. Ferrucci, "Introduction to "This is Watson"," in *IBM Journal of Research and Development*, vol. 56, no. 3.4, pp. 1:1-1:15, May-June 2012, doi: 10.1147/JRD.2012.2184356. <https://ieeexplore.ieee.org/document/6177724>
- [2] G. Tesauro, D. C. Gondek, J. Lenchner, J. Fan and J. M. Prager, "Simulation, learning, and optimization techniques in Watson's game strategies," in *IBM Journal of Research and Development*, vol. 56, no. 3.4, pp. 16:1-16:11, May-June 2012, doi: 10.1147/JRD.2012.2188931. <https://arxiv.org/ftp/arxiv/papers/1402/1402.0571.pdf>
- [3] [Strategies and skills of Jeopardy! champions - Wikipedia](#)
- [4] [J! Archive - Show #8276, aired 2020-11-09](#)
- [5] [Jeopardy! Round Daily Double statistics for Season 37](#)